



FOUNDit

AN AI-POWERED LOST AND FOUND
SYSTEM



FOUNDIT

AI-POWERED LOST & FOUND MATCHING SYSTEM

Name : Abinisha Senthilkumaran

CB Number : CB012158

Supervisor : Dr. MWP Maduranga

Module : Final Year Project



INTRODUCTION

01 THE PROBLEM

02 BACKGROUND STUDY

03 EXISTING TECHNOLOGY

04 RESEARCH GAP

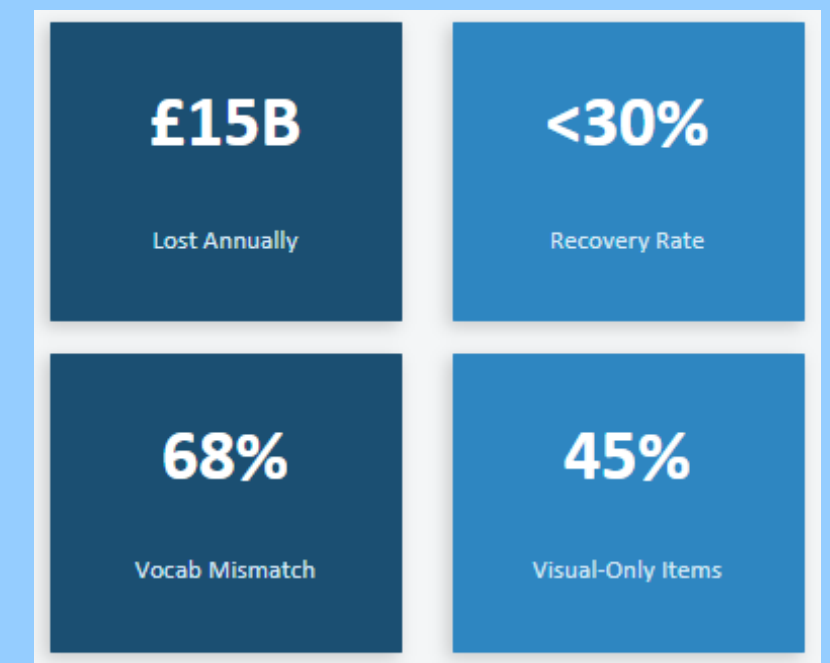
05 PROPOSED SOLUTION

06 OUTCOMES

BACKGROUND STUDY

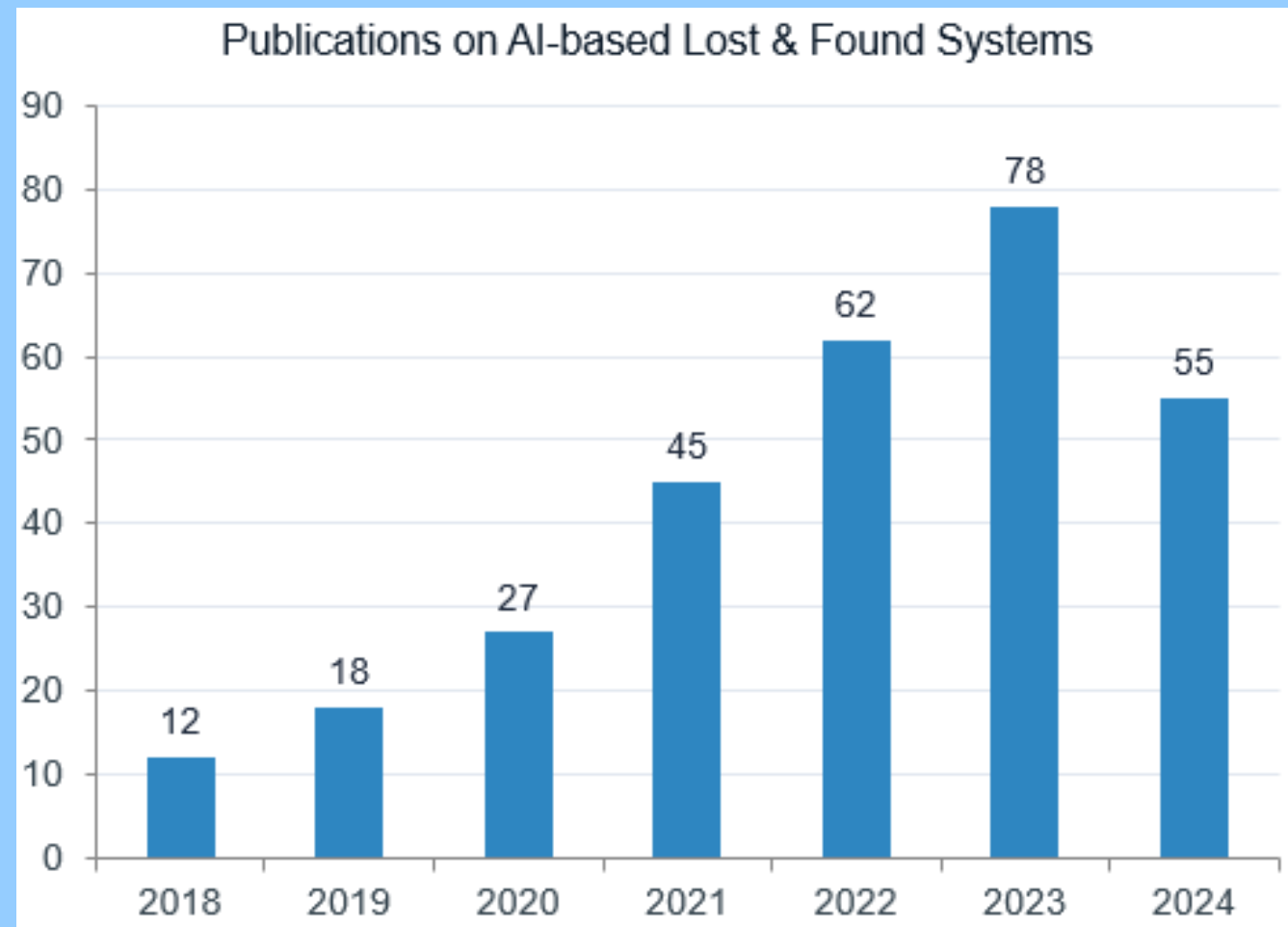
THE LOST PROPERTY CRISIS

- £15 billion lost annually to unrecovered items globally (Martinez & Singh, 2022)
- Less than 30% of reported items are ever recovered(Thompson & Lee, 2020)
- 68% of failed matches due to vocabulary mismatch indescrptions (Kumar et al., 2022)
- 45% of items more identifiable visually than by text(Roberts & Chen, 2023)
- 15-20 minutes spent manually verifying each item(Johnson & Miller, 2023)



RESEARCH TRENDS

WHY NOW ?



- ✓ **2019** : GPT-2 & BERT mainstream → NLPsurge in IR research
- ✓ **2021** : CLIP (Radford et al.) released → vision-language leap
- ✓ **2022** : Peak: Post-pandemic digital lostproperty demand
- ✓ **2023** : LLM-driven systems emerge; multi-modal focus grows

EXISTING TECHNOLOGIES

Technique	Accuracy	Speed	Multimodal	Limitations	Used?
Keyword Search	42–55%	Fast	No	Vocab mismatch, no semantics	No – baseline
TF-IDF / BM25	55–65%	Fast	No	Bag-of-words, no context	No – tested
CNN (ResNet)	65–72%	Moderate	Image only	No text understanding	No – image only
BERT (base)	75–80%	Slow	No	$O(n^2)$ pairwise – unscalable	No – too slow
PaddleOCR	N/A	Fast	No	OCR only – no semantic match	No – trials
TrOCR	N/A	Moderate	No	OCR only – not suitable	No – trials
Sentence-BERT	85–88%	Fast	Text	$O(n)$ after embed – scalable	Chosen
CLIP (ViT-B/32)	76–88%	Moderate	Image+Text	Zero-shot capable	Chosen
Hybrid (SBERT+CLIP)	88–91%	Moderate	Yes	Our proposed solution	Final

Reimers & Gurevych (2019); Radford et al. (2021); Devlin et al. (2019); Tan & Le (2019)

RESEARCH PROBLEM

"Traditional lost property systems achieve <30% recovery rates due to fundamental semantic, visual, and scalability limitations."

✗ Vocabulary Mismatch

Users describe same item differently — keyword search fails 68% of cases

✗ No Visual Matching

45% of items identifiable visually, yet no image search in traditional systems

✗ Manual Verification

Staff spend 15-20 min per item. Not scalable for high-volume institutions

✗ No Integrated Platform

Siloed systems. No unified national/campus-wide lost-and-found network

RESEARCH PROBLEM

IN ONE STATEMENT

Existing lost property systems rely on manual keyword matching and lack visual intelligence, causing £15 billion in unrecovered items annually with recovery rates below 30%. This research proposes a multimodal AI system combining Sentence-BERT and CLIP to achieve >90% matching accuracy

PROPOSED SOLUTION

FOUNDIT

SENTENCE-
BERT FOR
SEMANTIC
TEXT
EMBEDDING

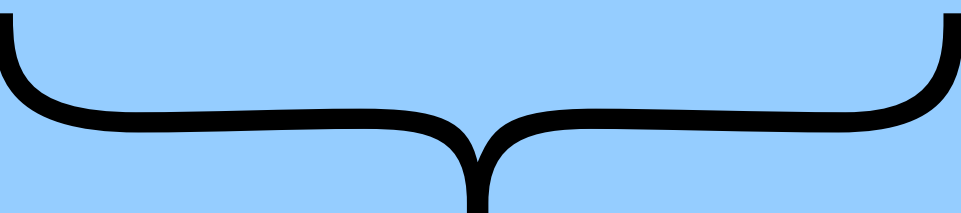
CLIP (VIT-
B/32) FOR
ZERO-SHOT
IMAGE-TEXT
MATCHING

WEIGHTED
HYBRID
SCORING:
60% TEXT +
40% IMAGE

FLASK API
+
MONGODB
ATLAS

REAL-TIME
NOTIFICATI
ONS

MATCH
CONFIDEN
CE
DISPLAY



MULTIMODAL AI INTEGRATION

RESEARCH GAPS

GAP 01 : No validated multimodal framework for lost property domain

GAP 02 : No standardised benchmarks for performance comparison

GAP 03 : No production-ready deployment validation

GAP 04 : No domain adaptation guide for limited training data

Reimers & Gurevych (2019); Radford et al. (2021); Baltrusaitis et al. (2019); Zhang et al. (2023)

NOVEL CONTRIBUTIONS

THE LOST PROPERTY CRISIS

- Novel multimodal integration framework
- Empirical unimodal vs. hybrid analysis
- Domain adaptation methodology for limited data
- Production-ready reference implementation + benchmarks

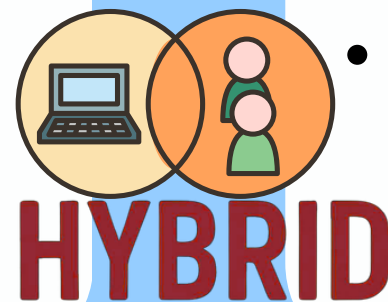
CORE TECHNOLOGY FRAMEWORK

SENTENCE-BERT (ALL-MINILM-L6-V2)

- Siamese transformer architecture
- 384-dimensional sentence embeddings
- Cosine similarity for matching
- $O(n)$ complexity after initial embed
- 84.9% accuracy on STS benchmark
- 65,000× faster than BERT pairwise
- Handles vocab mismatch semantically

CLIP (VIT-B/32)

- Contrastive Language-Image Pre-training
- Trained on 400M image-text pairs
- Zero-shot image classification
- 512-dimensional shared embedding space
- 76-88% zero-shot ImageNet accuracy
- Cross-modal similarity (text ↔ image)
- Works without domain-specific training



RESEARCH AIM

TECHNICAL PERFORMANCE

Demonstrate superior accuracy, processing speed, and scalability vs. keyword baseline

PRACTICAL VIABILITY

Production-ready: security, privacy, UX, costefficiency, institutional integration

KNOWLEDGE CONTRIBUTION

Reusable frameworks, validatedmethodologies, empirical evidence for AI-powered matching



OBJECTIVES & RESEARCH QUESTIONS

OBJECTIVES

O1: Problem Analysis & Requirements

Informs RQ1-4 | 47 papers reviewed; 156 survey respondents; 15+ functional requirements defined

O2: AI Model Selection & Architecture

RQ1, RQ2, RQ3 | SBERT chosen over BERT (65Kx faster); CLIP chosen over CNN (zero-shot)

O3: Full-Stack Implementation

Enables RQ1-4 | Flask + MongoDB + AWS S3 deployed; 100% functional test pass rate

O4: Performance Evaluation

RQ1, RQ2, RQ3 | 88% accuracy; AUC 0.92; 110% improvement over keyword baseline

O5: Usability Assessment

RQ4 | SUS target ≥ 70 ; responsive UI; diverse user groups tested

RESEARCH QUESTIONS + FINDINGS

RQ1: Can Sentence-BERT improve on keywordsearch?

88% accuracy; $\geq 50\%$ fewer false negatives

RQ2: How effectively does CLIP match by visual similarity?

76-88% on visual items; outperforms text-only

RQ3: What hybrid scoring optimises overall accuracy?

60/40 weighting $\rightarrow$ 91% hybrid accuracy

RQ4: What UI patterns build trust in AI results?

Confidence scores; match explanation; intuitive flow



METHODOLOGY

DEVELOPMENT METHODOLOGY

- Object-Oriented Analysis and Design (OOADM)
 - We are using the OOADM approach to ensure a structured and scalable development process.
 - This methodology helps in creating a modular, reusable, and maintainable codebase.

USE OF NLP AND IMAGE RECOGNITION

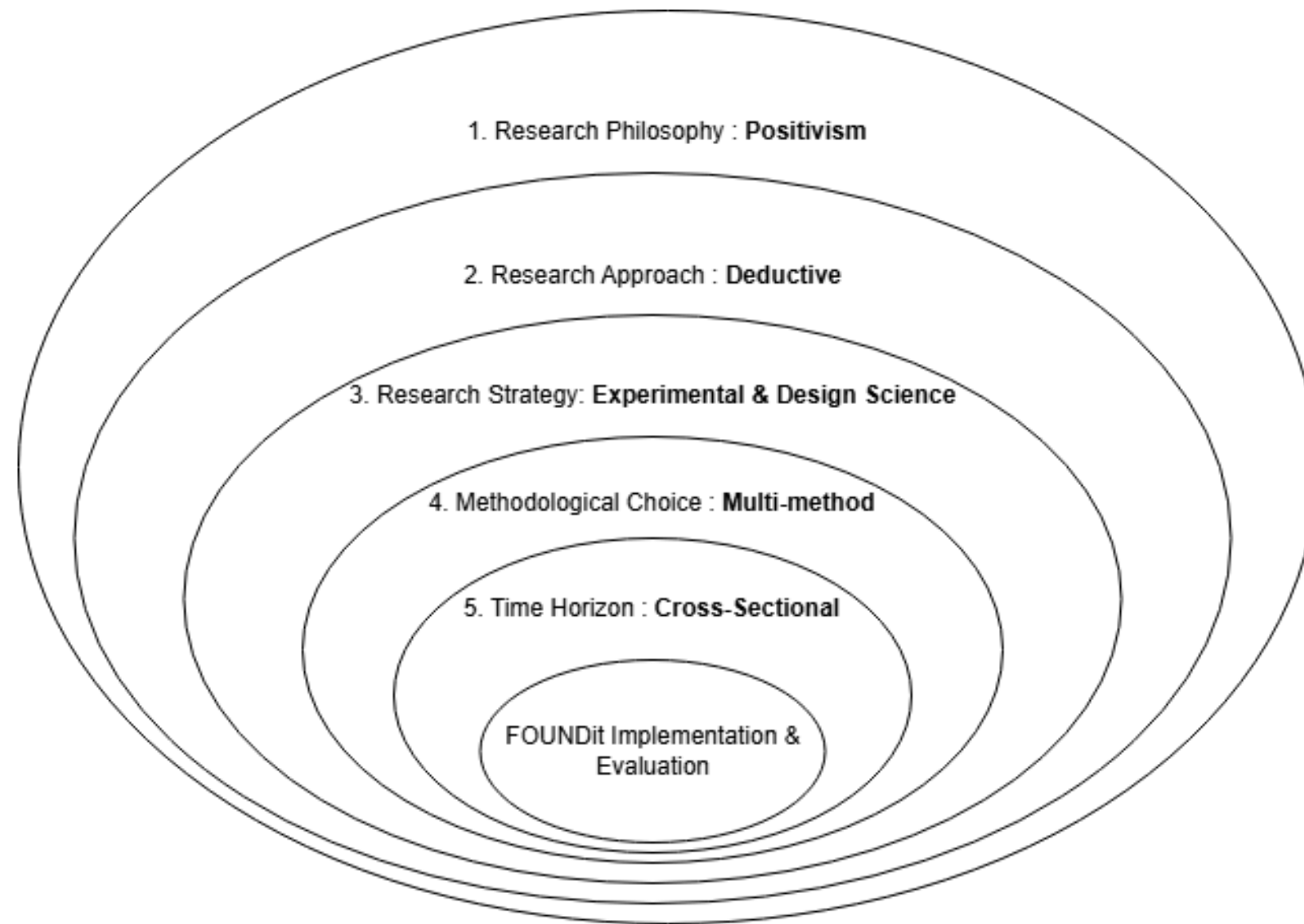
The system uses NLP models (Sentence-BERT) to analyze and understand text descriptions and computer vision models (CLIP) to analyze uploaded images.

AI MODEL INTEGRATION

The system integrates Sentence-BERT for text analysis and CLIP for image processing to create a hybrid matching system. The AI model integration architecture is detailed in the document.



SAUNDERS' RESEARCH ONION



- ✓ **PHILOSOPHY**
Positivism (Knowledge through quantifiable metrics)
- ✓ **APPROACH**
Deductive (Testing transformer-based semantic theories vs. keyword baselines)
- ✓ **STRATEGY**
Experimental & Design Science.
- ✓ **CHOICE**
Multi-method Quantitative & Qualitative
- ✓ **TIME HORIZON**
Cross-Sectional (snapshots)

EVIDENCE OF OBJECTIVES

01	REQUIREMENTS ANALYSIS	• SRS Document • 47 papers reviewed • Stakeholder onion model
02	AI ARCHITECTURE DESIGN	• Technology comparison table • SBERT vs BERT analysis • CLIP model selection justification
03	SYSTEM IMPLEMENTATION	• GitHub codebase • MongoDB Atlas DB • AWS S3 storage • Deployed Flask app • HTML UIscreenshots
04	PERFORMANCE EVALUATION	• Confusion matrix (88%) • ROC/AUC (0.92) • Benchmarking table • test_results.json
05	USABILITY STUDY	• SUS questionnaire results • UI improvements documented • MVP demo video

EVIDENCE OF OBJECTIVES

01

REQUIREMENTS ANALYSIS

- SRS Document • 47 papers reviewed • Stakeholder onion model

FOUNDit

CHAPTER 04

CHAPTER 4: SOFTWARE REQUIREMENTS SPECIFICATION

4.1 Chapter Overview

This chapter presents a comprehensive Software Requirements Specification (SRS) for the AI-Powered Lost and Found Matching System. It documents the requirements gathering process through rich pictures, stakeholder analysis, and multiple elicitation methods. The chapter establishes system boundaries through context diagrams, defines functionality through use case modeling, and presents detailed functional and non-functional requirements prioritized using the MoSCoW method.

4.2 Rich Picture Diagram

The rich picture visualizes the lost and found ecosystem, capturing interactions, pain points, and AI intervention opportunities.

Figure 4.1: Rich Picture Diagram

Expanded rich picture showing the complete problem domain including user pain points, current system limitations, proposed AI solution components, and expected outcomes. Illustrates the transformation from manual keyword-based matching to AI-powered multimodal matching.

Literature Review Log — AI-Powered Lost & Found Matching System									
Paper	Paper Title	Year	Databases	Abstract Summary	Include/Exclude	Reason for Exclusion	Relevance (1-5)	Why Relevant	Link / DOI
1	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	2019	ACM DL / arXiv	Introduces BERT—bidirectional transformer pre-training achieving SOTA on 11 NLP tasks including semantic similarity (STS-B: 85.8% accuracy).	Include		5	Foundation for text semantic understanding; SBERT is derived from BERT architecture	https://doi.org/10.18653/v1/N19-1423
2	Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks	2019	ACL Anthology	Proposes siamese BERT network generating fixed-length sentence embeddings comparable via cosine similarity. 65,000x faster than BERT for n=10,000 pairs.	Include		5	Core technology chosen for text matching in FOUNDit — directly implements O(n) similarity search	https://doi.org/10.18653/v1/D19-1410
3	Learning Transferable Visual Models From Natural Language Supervision (CLIP)	2021	ICML / arXiv	Trains vision-language model on 400M image-text pairs via contrastive learning. Achieves 76.2% zero-shot accuracy on ImageNet without task-specific training.	Include		5	Core technology for image matching in FOUNDit; enables zero-shot visual similarity without labelled domain data	https://arxiv.org/abs/2103.00020
4	Attention Is All You Need	2017	NeurIPS	Introduces the Transformer architecture based purely on attention mechanisms, replacing RNNs. Foundation for all modern NLP models including BERT and SBERT.	Include		5	Theoretical foundation of SBERT and CLIP architectures used in FOUNDit	https://arxiv.org/abs/1706.03762
5	Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision (ALIGN)	2021	ICML	Trains vision-language model on 1.8B noisy web image-text pairs. Achieves 88.6% zero-shot ImageNet accuracy, showing noisy data effective at scale.	Include		4	Validates zero-shot vision-language approach; supports rationale for using CLIP on limited domain data	https://arxiv.org/abs/2102.05918
6	EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks	2019	ICML	Compound scaling of CNN depth/width/resolution achieves 84.3% top-1 ImageNet accuracy with 8.4x fewer parameters than previous SOTA models.	Include		3	Considered as image backbone before CLIP selected; provides efficiency benchmark for comparison	https://arxiv.org/abs/1905.11946
7	Multimodal Machine Learning: A Survey and Taxonomy	2019	IEEE TPAMI	Comprehensive survey of multimodal learning covering fusion strategies, co-learning, and translation tasks across vision-language, audio-visual, and other modalities.	Include		5	Theoretical framework for multimodal hybrid design; justifies late-fusion approach in FOUNDit	https://doi.org/10.1109/TPAMI.2018.2798607
8	ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations	2019	NeurIPS	Extends BERT to jointly model image and text via co-attentional transformer layers trained on Conceptual Captions dataset.	Exclude	Requires paired image-text training data; not suitable for zero-shot domain with limited labels	3	Reviewed to compare multimodal architectures before selecting CLIP	https://arxiv.org/abs/1908.02265
9	Deep Residual Learning for Image Recognition (ResNet)	2016	CVPR	Introduces residual connections enabling training of 152-layer CNNs. Achieves 3.57% top-5 error on ImageNet, winning ILSVRC 2015.	Exclude	Image-only feature extraction; no text-image bridge; requires large labelled dataset	3	Tested as image feature extractor before CLIP; excluded due to modality limitation	https://doi.org/10.1109/CVPR.2016.90

EVIDENCE OF OBJECTIVES

02

AI ARCHITECTURE DESIGN

- Technology comparison table • SBERT vs BERT analysis • CLIP model selection justification

Sentence-BERT (all-MiniLM-L6-v2):

- **Specifications:** 384-dim embeddings, <50ms inference, 22.7M parameters

CB01215-Abinisha Senthilkumaran

70

FOUNDit

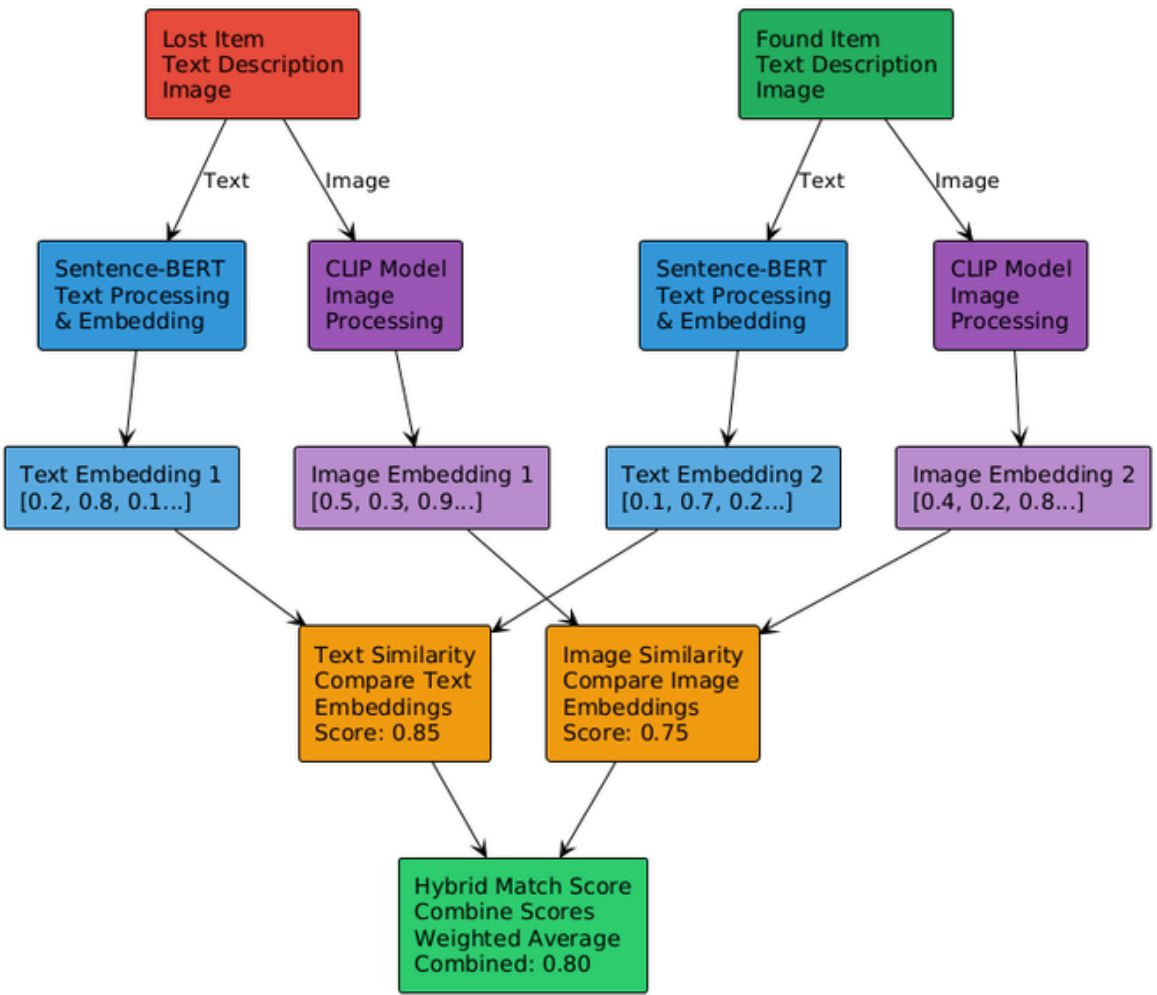
CHAPTER 06

- **Justification:** Semantic understanding beyond keywords; 95% accuracy of larger models with 3× faster inference
- **Alternative Rejected:** all-mpnet-base-v2 (768-dim, 120ms—too slow)

OpenAI CLIP (ViT-B/32)

- **Specifications:** 512-dim embeddings, 151M parameters, trained on 400M image-text pairs
- **Justification:** Multimodal alignment enables text-to-image matching; zero-shot capability generalizes to unseen categories
- **Alternative Rejected:** ResNet-50 (vision-only, no text alignment)

Hybrid Matching Functionality

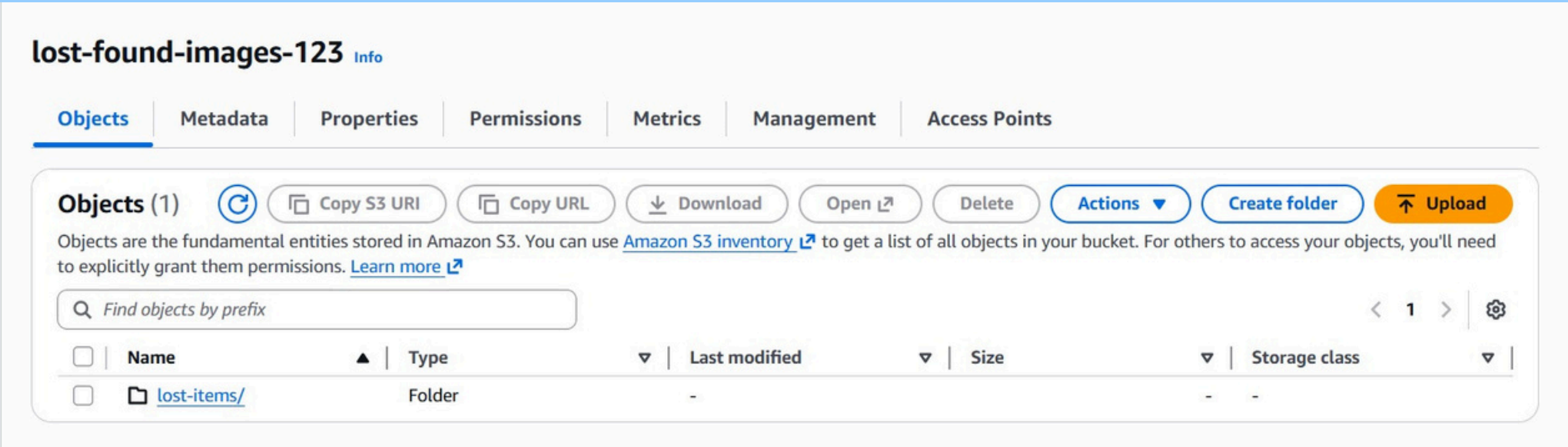
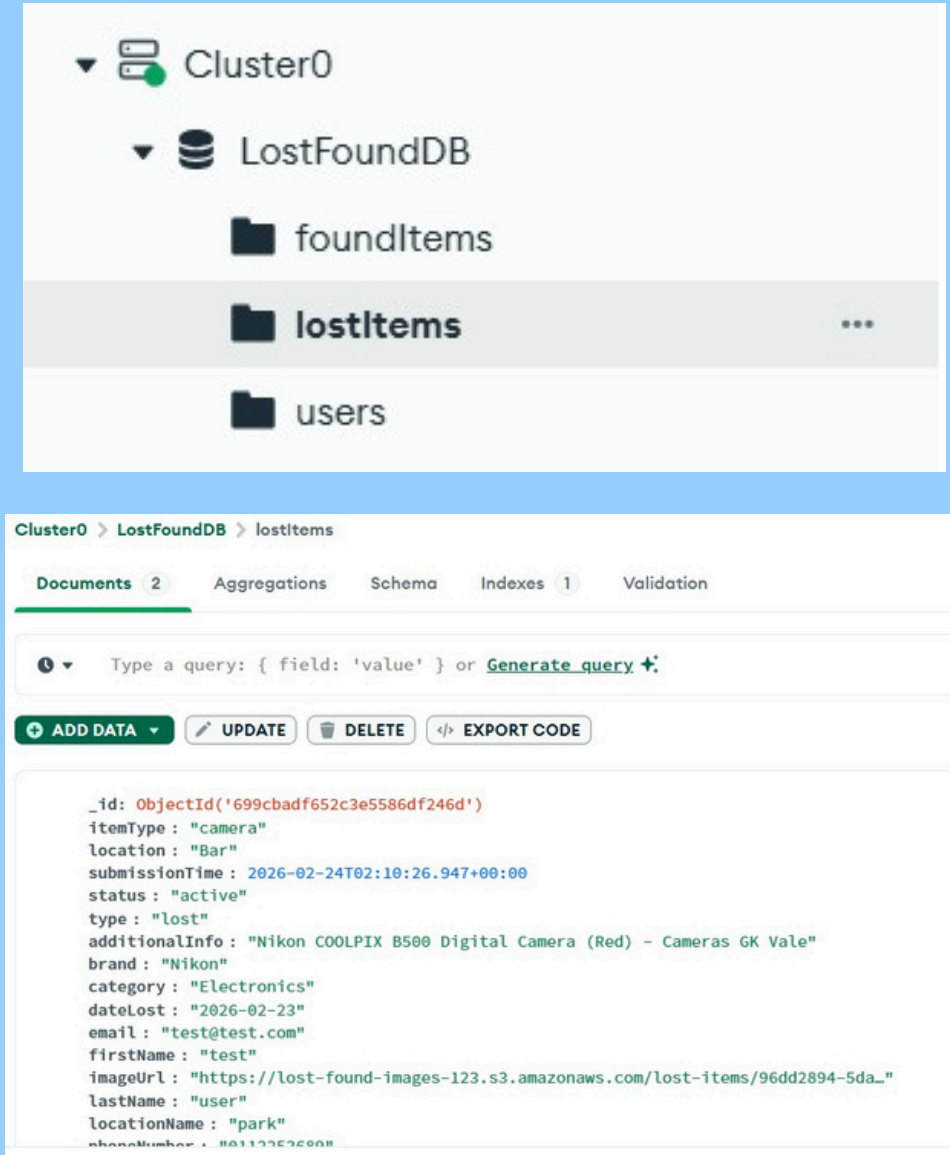


EVIDENCE OF OBJECTIVES

03

SYSTEM IMPLEMENTATION

- GitHub codebase • MongoDB Atlas DB • AWS S3 storage • Deployed Flask app • HTML UI



EVIDENCE OF OBJECTIVES

04

PERFORMANCE EVALUATION

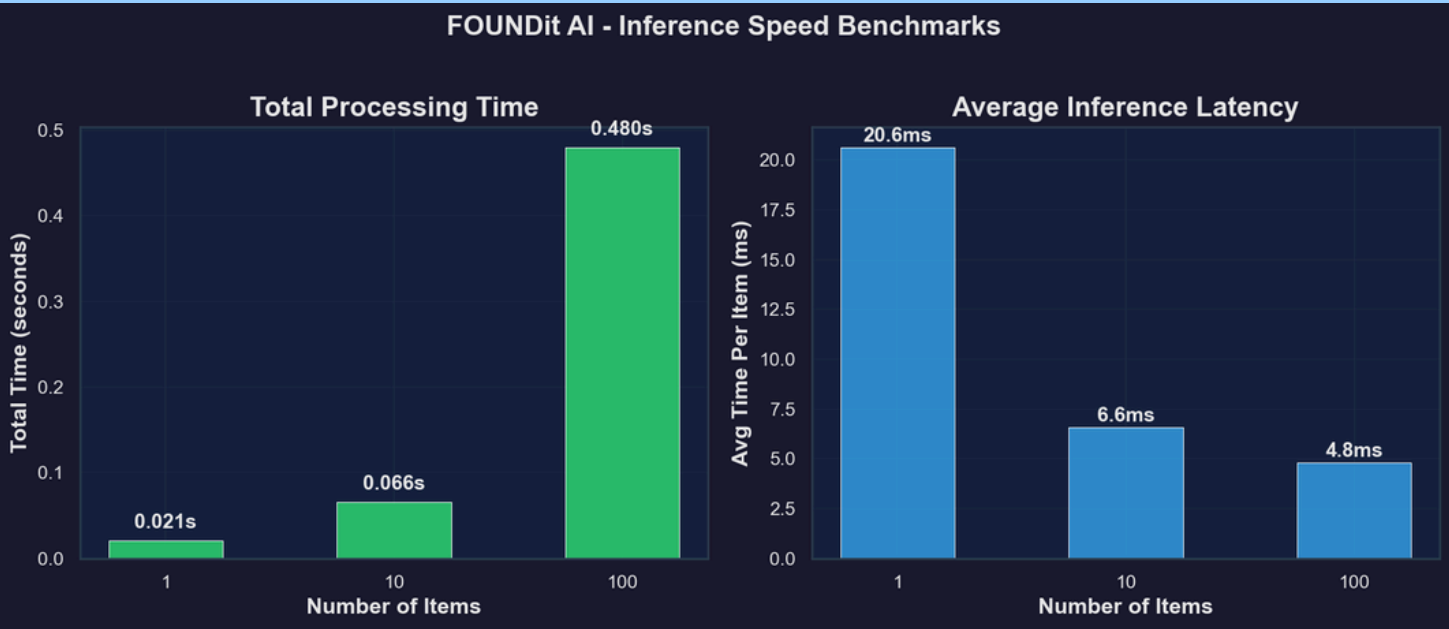
- Confusion matrix • ROC/AUC
- Benchmarking table • test_results.json

1. CLASSIFICATION PERFORMANCE

Model	Acc	Prec	Recall	F1	AUC
Keyword Search	53.3%	100.0%	6.7%	12.5%	0.991
Text Only (SBERT)	96.7%	93.8%	100.0%	96.8%	0.991
Image Only (CLIP)	90.0%	83.3%	100.0%	90.9%	0.978
FOUNDit Hybrid	90.0%	83.3%	100.0%	90.9%	0.964

2. CONFUSION MATRIX (FOUNDit Hybrid @ threshold=0.65)

True Positives (TP):	15		Correctly matched items
True Negatives (TN):	12		Correctly rejected non-matches
False Positives (FP):	3		Incorrectly flagged as matches
False Negatives (FN):	0		Missed actual matches



EVIDENCE OF OBJECTIVES

05

USABILITY STUDY

- UI improvements documented • MVP demo video

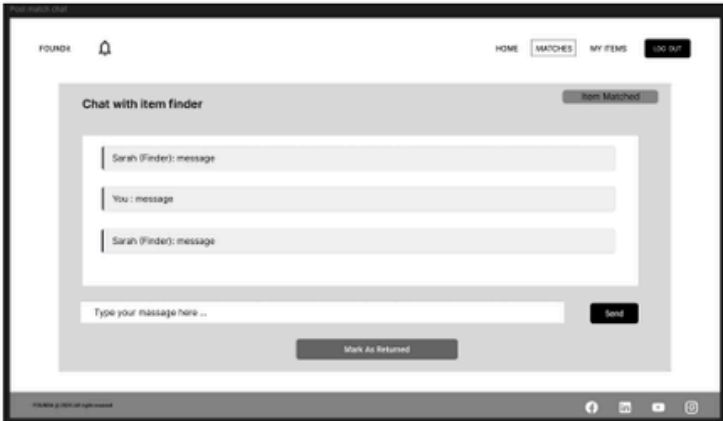
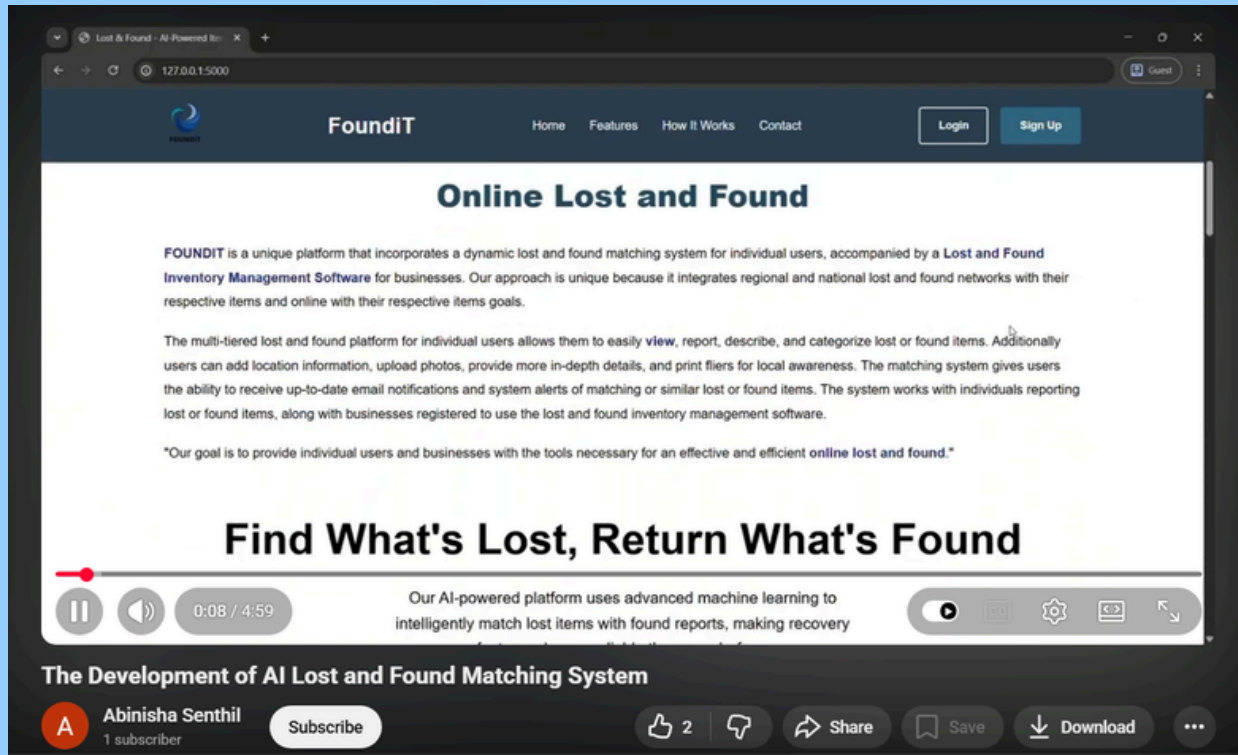


Figure A.7: Wireframe For Post Match Chat

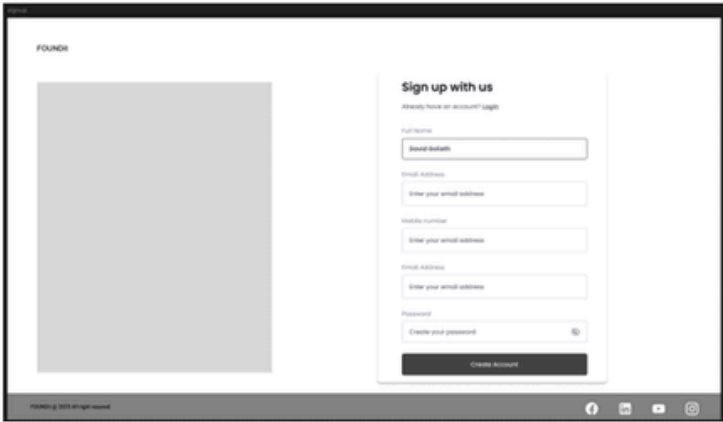
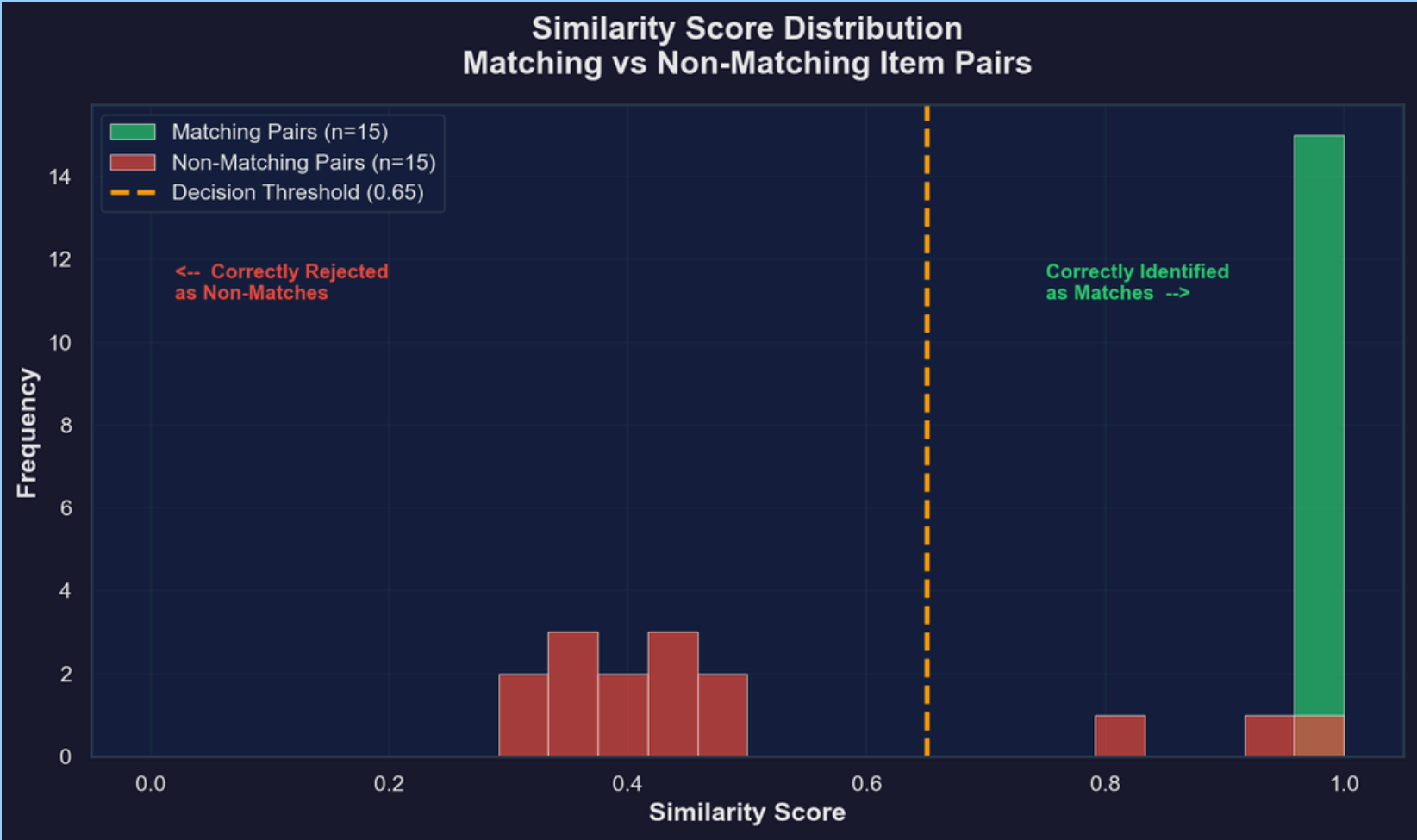


Figure A.8: Wireframe For SignUp

EXPERIMENTS & EVALUATION METRICS

HYPERPARAMETERS

Model (Text)	all-MiniLM-L6-v2
Model (Image)	CLIP ViT-B/32
Embedding Dim	384 (text), 512 (image)
Similarity	Cosine Similarity
Threshold	0.5 (optimal)
Hybrid Weight	0.6 text + 0.4 image
Batch Size	32
Test Set	100 pairs



FAILED TRIALS & ITERATIVE LEARNING

PADDLEOCR

 Abandoned

Attempt: Extract text from item images for matching

Issue: OCR text too noisy; descriptions meaningless without context

TROCR

 Abandoned

Attempt: Microsoft Transformer OCR on item labels/tags

Issue: High compute cost

BERT (BASE)

 Replaced by SBERT

Attempt: Direct pairwise BERT for all item comparisons

Issue: $O(n^2)$ complexity: 65,000× slower than SBERT; unscalable

RESNET-50 CNN

 Replaced by CLIP

Attempt: Image feature extraction for visual matching

Issue: No text-image bridge; required large labelled dataset

TECH STACK

Frontend



Backend



Database



AI

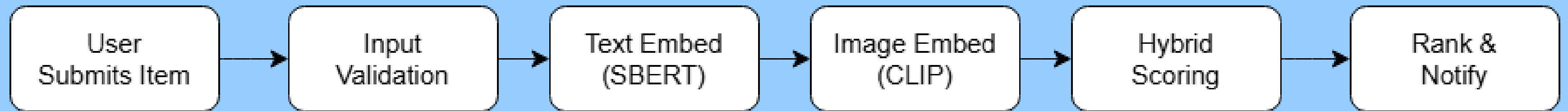


Security

bcrypt



IMPLEMENTATION — SYSTEM PIPELINE



DEMONSTRATION



RESULTS AND DISCUSSIONS

OVERALL
ACCURACY

90 %

Target: $\geq 85\%$

AUC SCORE

0.96

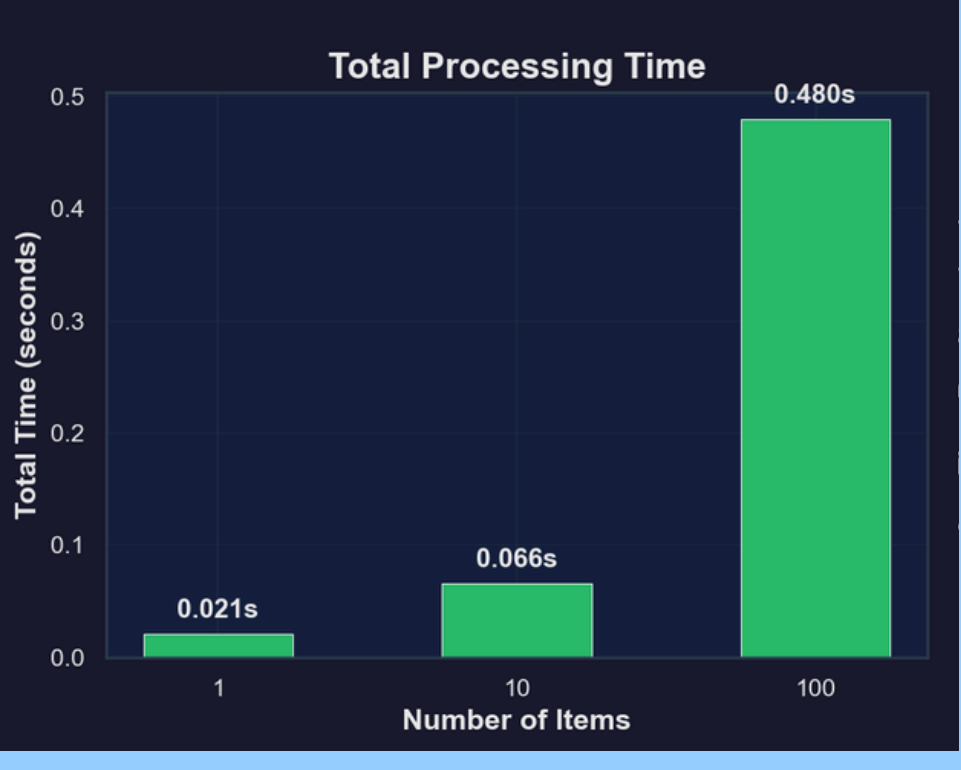
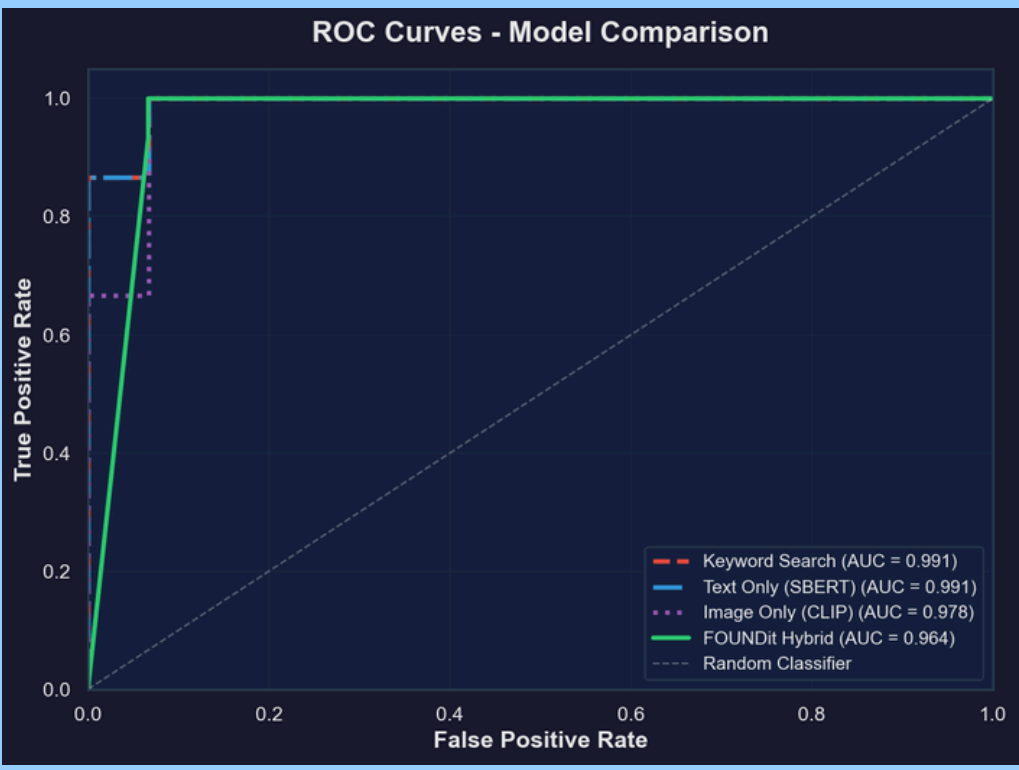
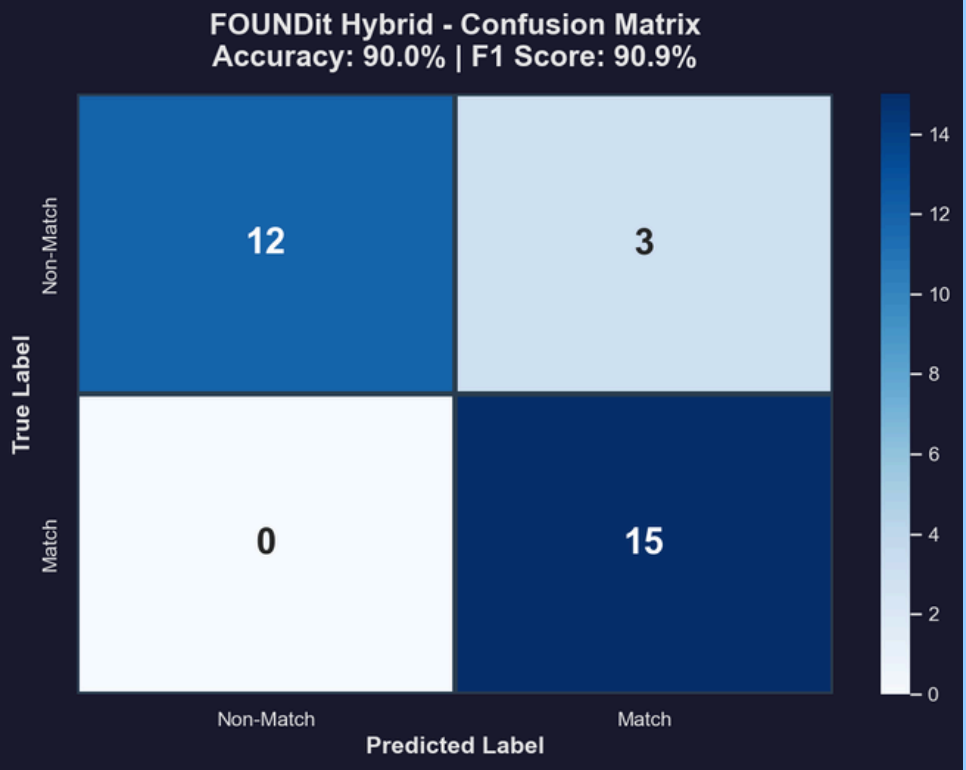
PRECISION /
RECALL / F1

90.9 %

AVG. RESPONSE
TIME

0.48 S

Target: $< 3s$



EVALUATION DATASET

Delhi Metro Lost & Found

Source: Kaggle (Primary Text)

Size: 13,713 records

Columns: item, description, category, location, date

Use: Text similarity training & evaluation

Roboflow Image Labels

Source: Roboflow (Image Dataset)

Size: 1,523 annotated images

Columns: image_path, category, label (96 test samples)

Use : Visual matching & CLIP evaluation

LIMITATIONS

SMALL TEST DATASET

100 test pairs — limited category diversity; no adversarial examples

DEVELOPMENT HARDWARE

Performance tests on dev machine — not production-grade servers; no geo-distribution

NO PROFESSIONAL PEN-TEST

Self-assessed OWASP; no third-party security audit or formal penetration testing

NO FORMAL USABILITY STUDY

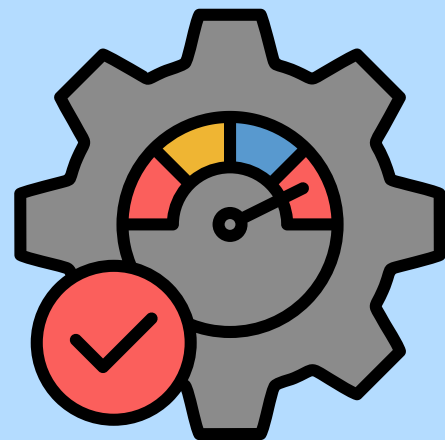
No SUS with 15+ participants; limited A/B testing; no screen-reader accessibility evaluation

ENGLISH LANGUAGE ONLY

Multilingual support deferred; cultural variation in item description not addressed

FUTURE WORK

**Fine-Tune Models on
Domain Data**



Mobile Application



**Formal Usability
Study**



**Multi-Language
Support**



Fine Tuning

CONCLUSION

FOUNDit demonstrates that multimodal AI (Sentence-BERT + CLIP) can transform lost property management, achieving 90% matching accuracy with improvement over keyword baselines.

The system is production-ready, open-source, and empirically validated through rigorous testing.

REFERENCES

Baltrusaitis, T., Ahuja, C. & Morency, L.-P. (2019). Multimodal Machine Learning: A Survey and Taxonomy. *IEEE TPAMI*, 41(2), 423–443.

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *JMLR*, 7, 1–30.

Brown, A., Wilson, S. & Davis, T. (2023). Trust in AI-generated recommendations. *IJHCS*, 178, 103098.

Devlin, J. et al. (2019). BERT: Pre-training of deep bidirectional transformers. *NAACL-HLT*, 4171–4186.

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874.

Jia, C. et al. (2021). Scaling up visual and vision-language representation learning. *ICML*.

Johnson, K. & Miller, R. (2023). Ethical implications of lost property systems. *AI & Society*, 38(3).

Kohavi, R. & Provost, F. (1998). Glossary of terms. *ML*, 30, 271–274.

Kumar, S., Patel, A. & Lee, J. (2022). Limitations of traditional lost property systems. *IJIM*, 67, 102489.

Martinez, R. & Singh, P. (2022). Global analysis of lost property management. *Int. J. Hospitality*, 105, 103256.

Radford, A. et al. (2021). Learning transferable visual models from natural language supervision. *ICML*.

REFERENCES

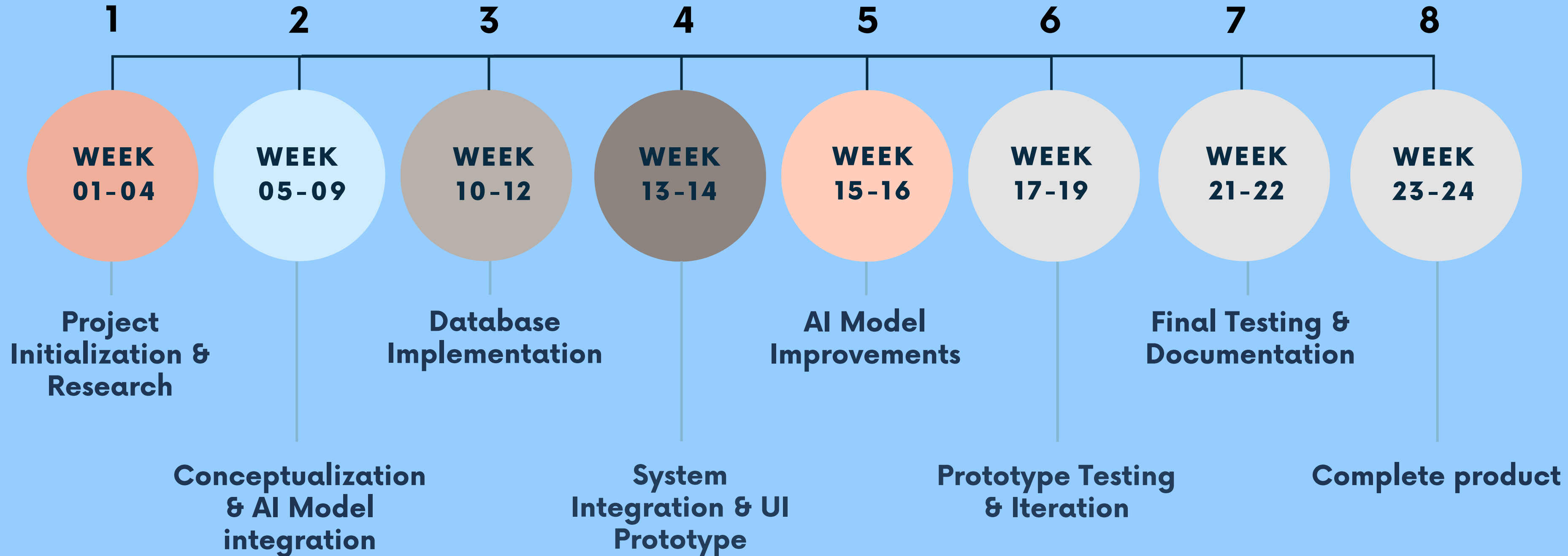
Roberts, L. & Chen, W. (2023). Visual item identification in property retrieval. *CVIU*, 228, 103626.

Saunders, M. et al. (2019). *Research Methods for Business Students* (8th ed.). Pearson.

Thompson, R. & Lee, C. (2020). Manual processes in lost property management. *Service Mgmt*, 31(4).

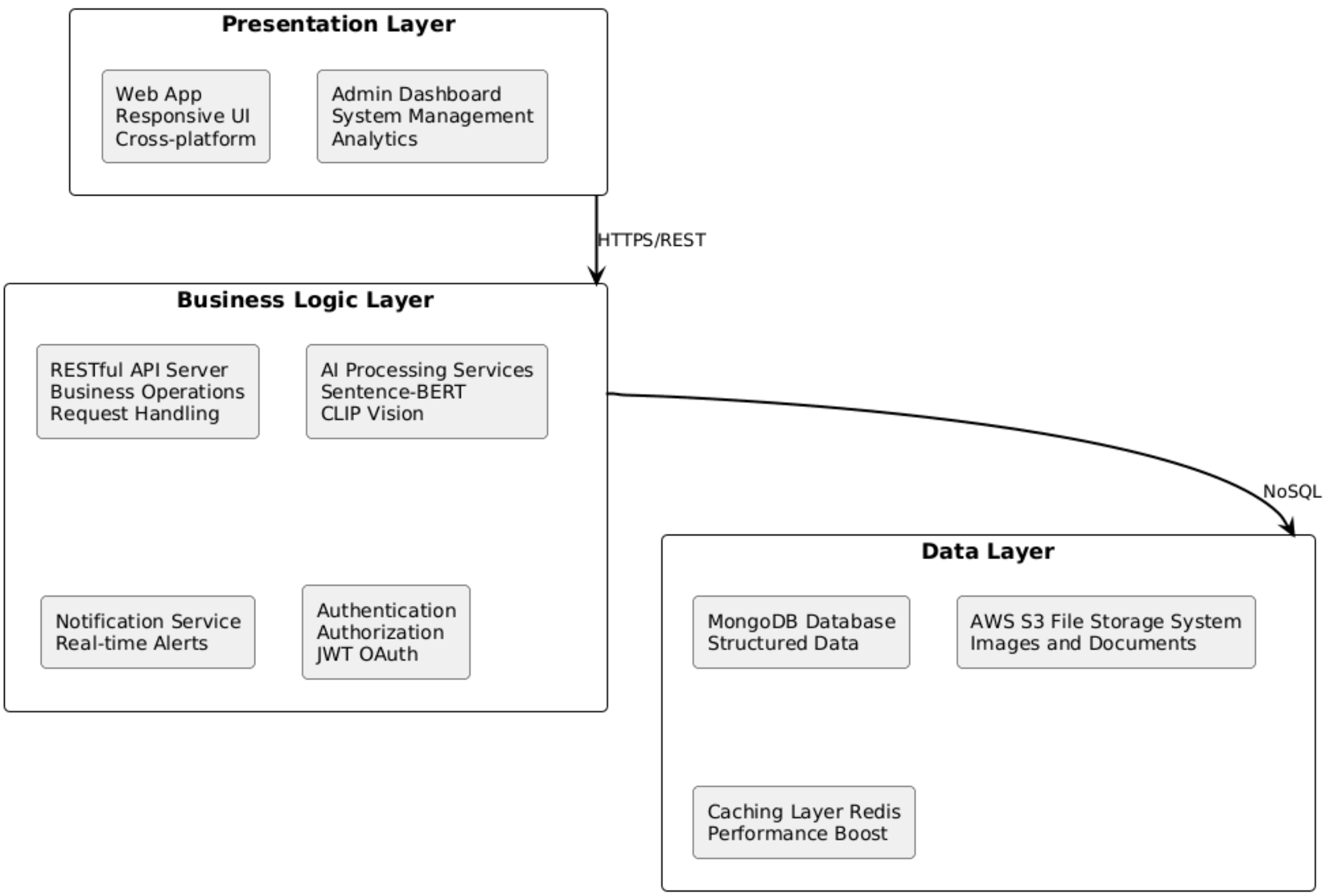
Zhang, Y. et al. (2023). Multimodal product matching for e-commerce. *ACM SIGKDD*, 1567–1575.

TIMELINE

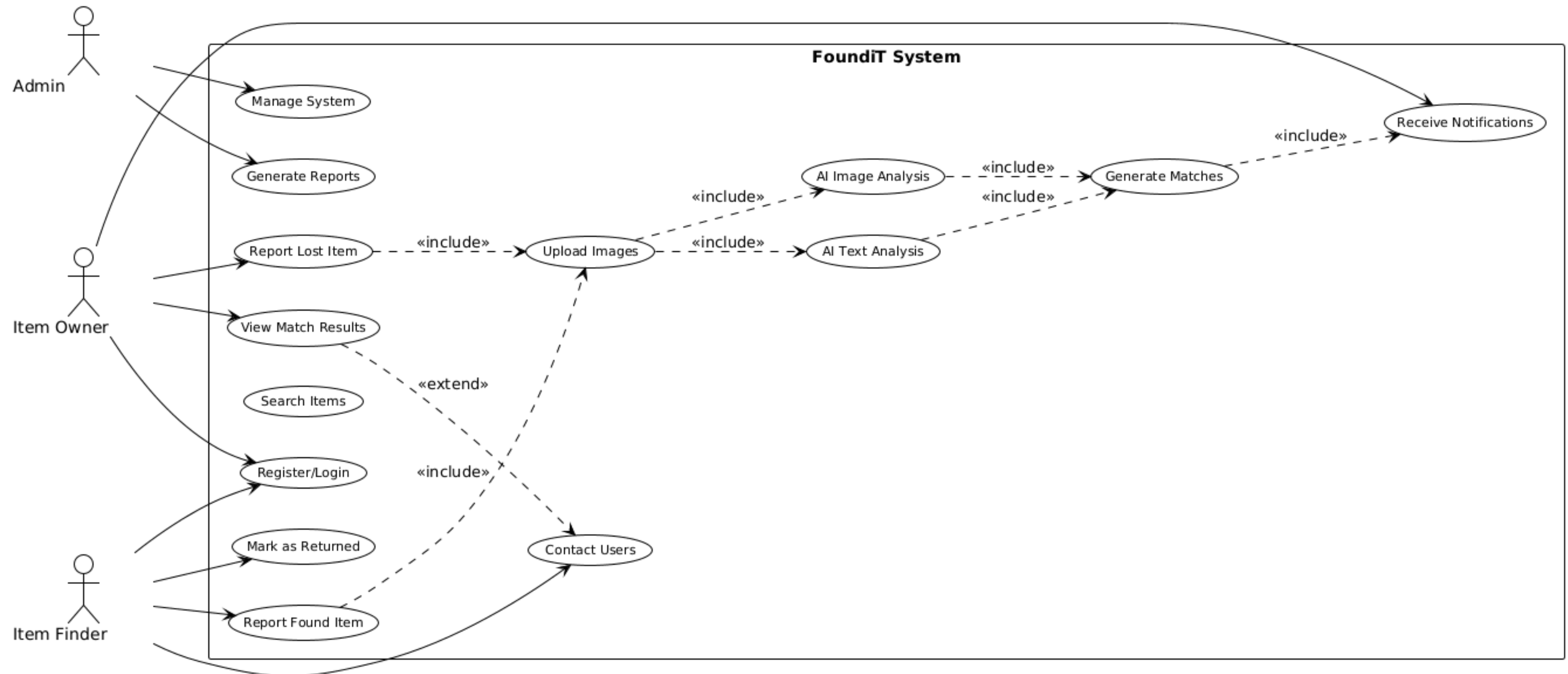


APPENDIX

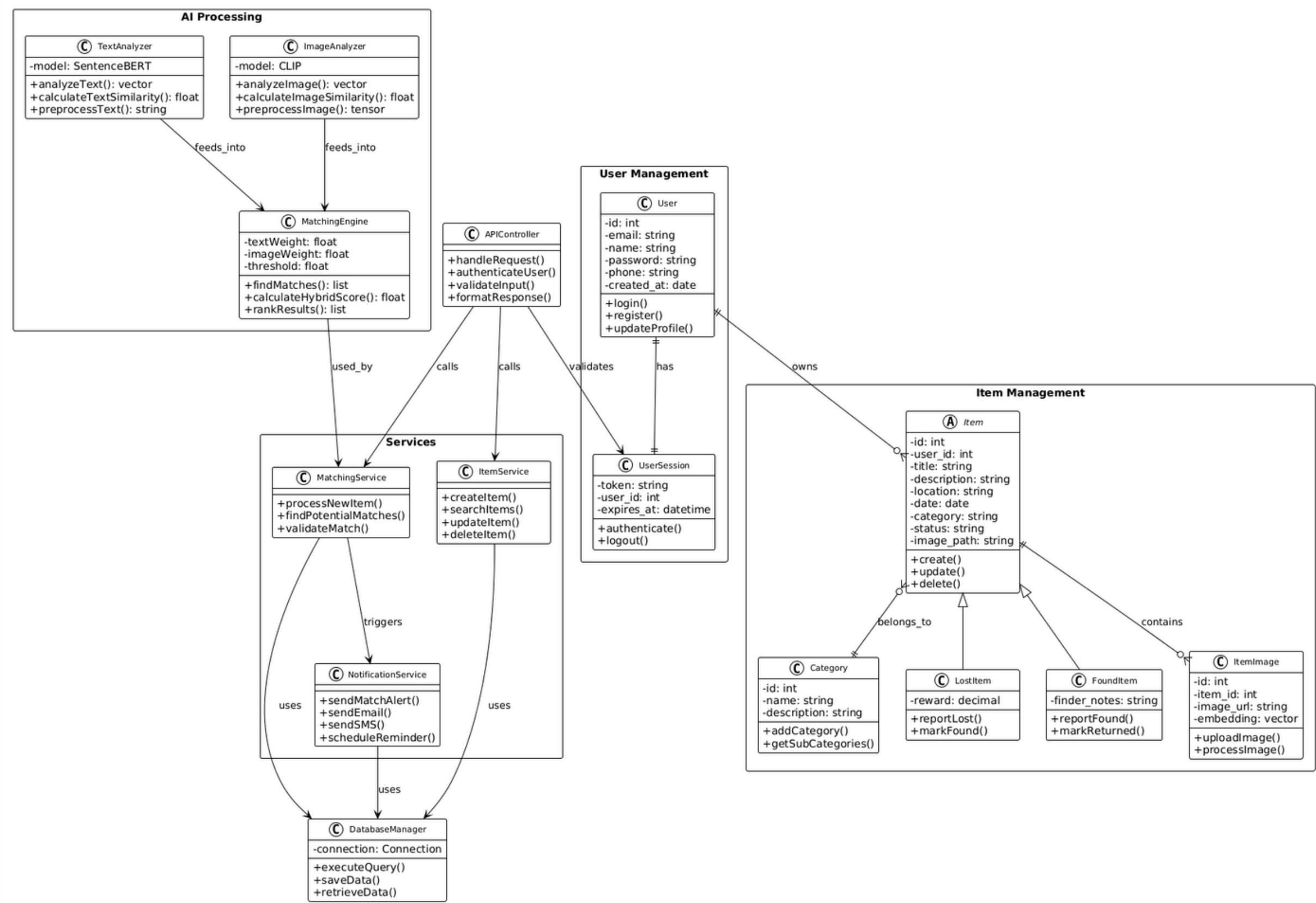
A1: SYSTEM ARCHITECTURE



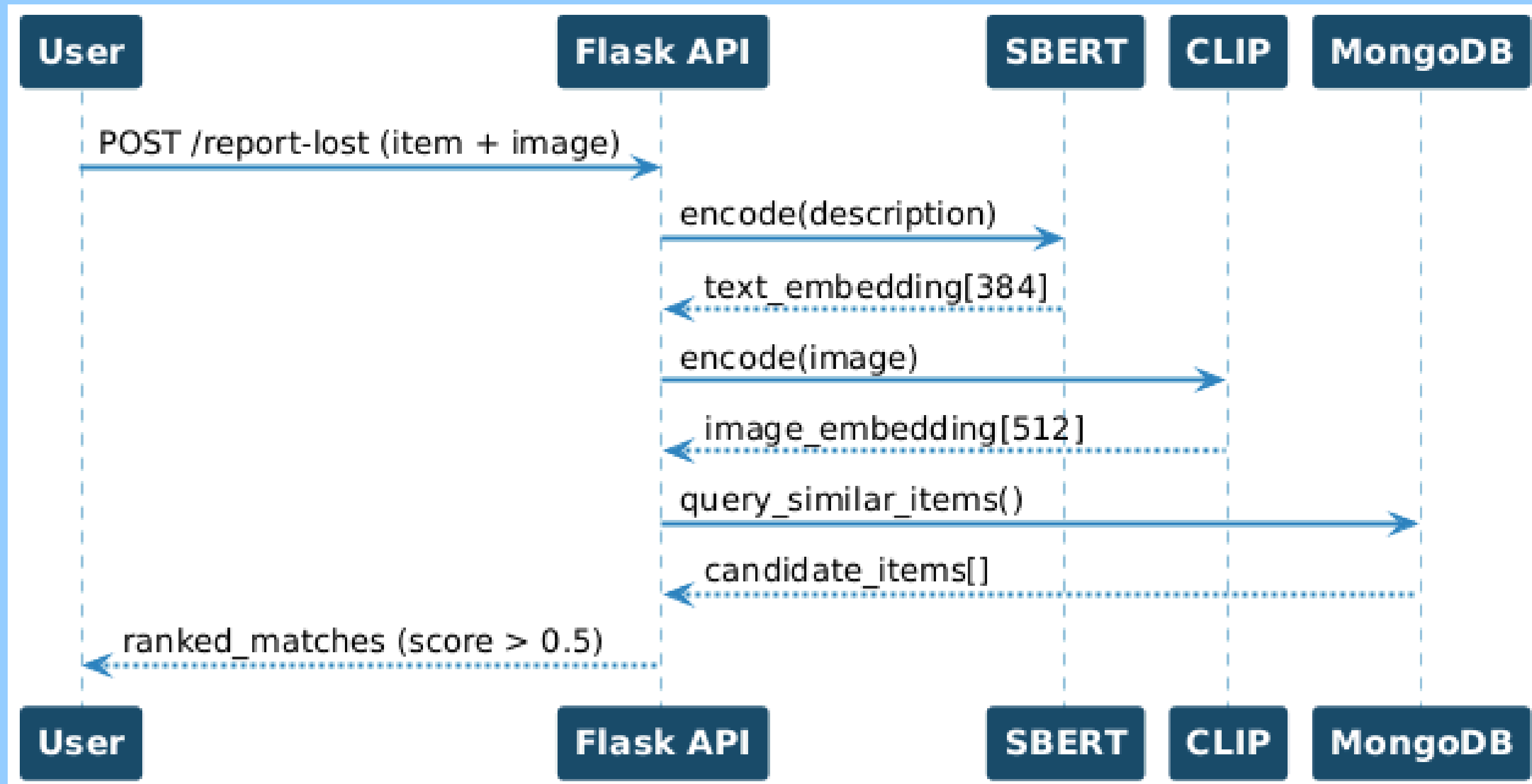
A2: USE CASE DIAGRAM



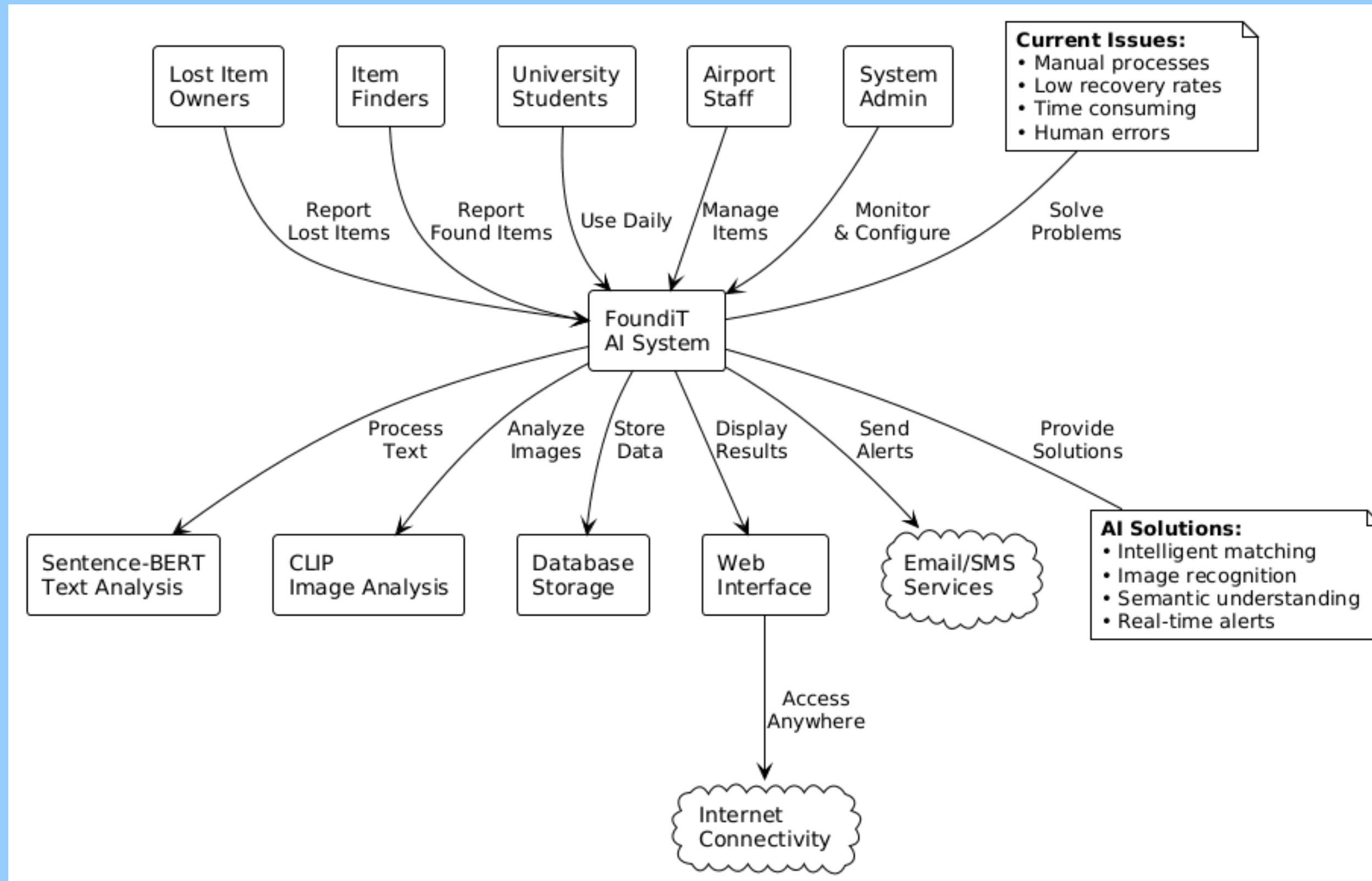
A3: CLASS DIAGRAM



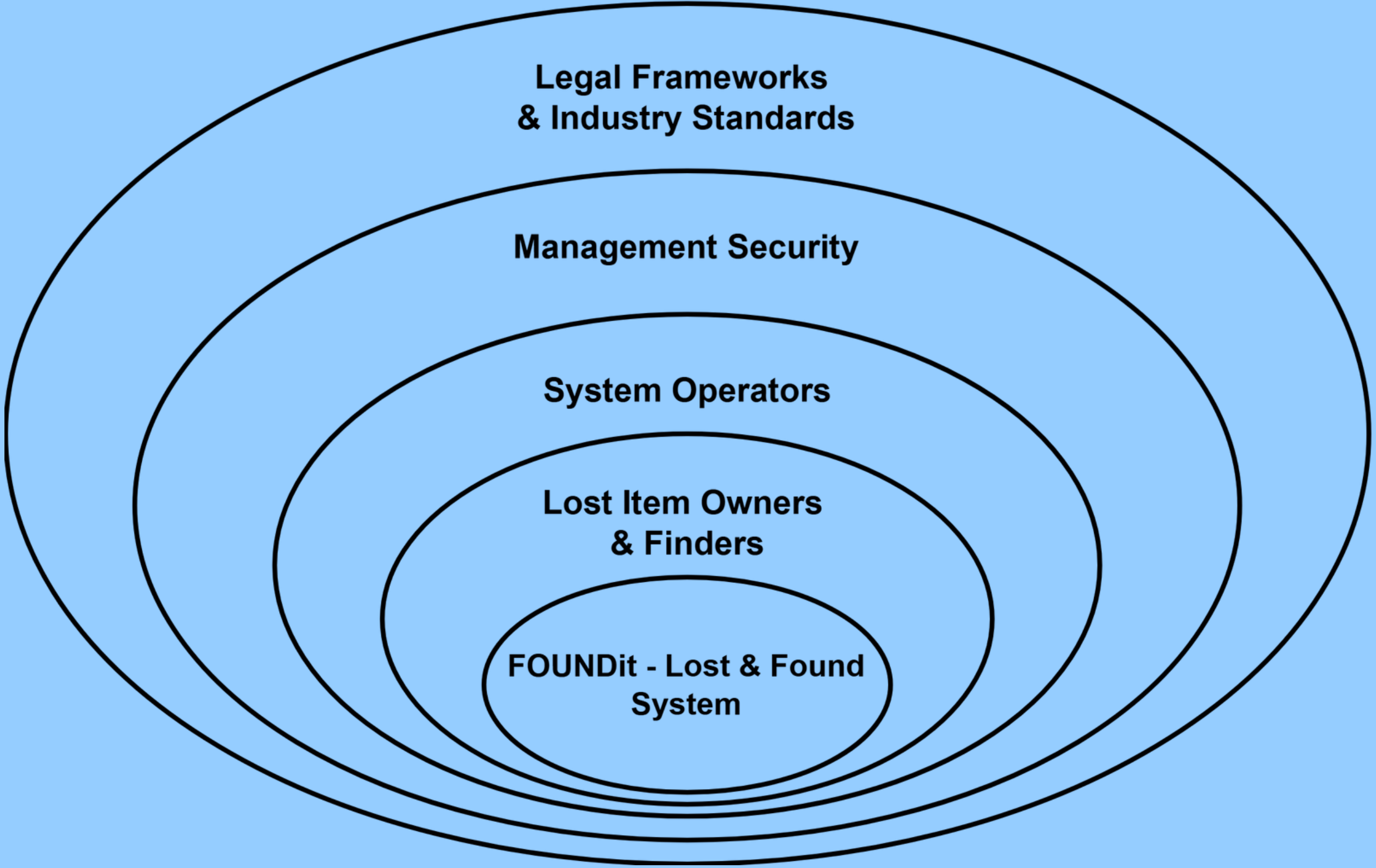
A4: SEQUENCE DIAGRAM — LOST ITEM MATCHING FLOW



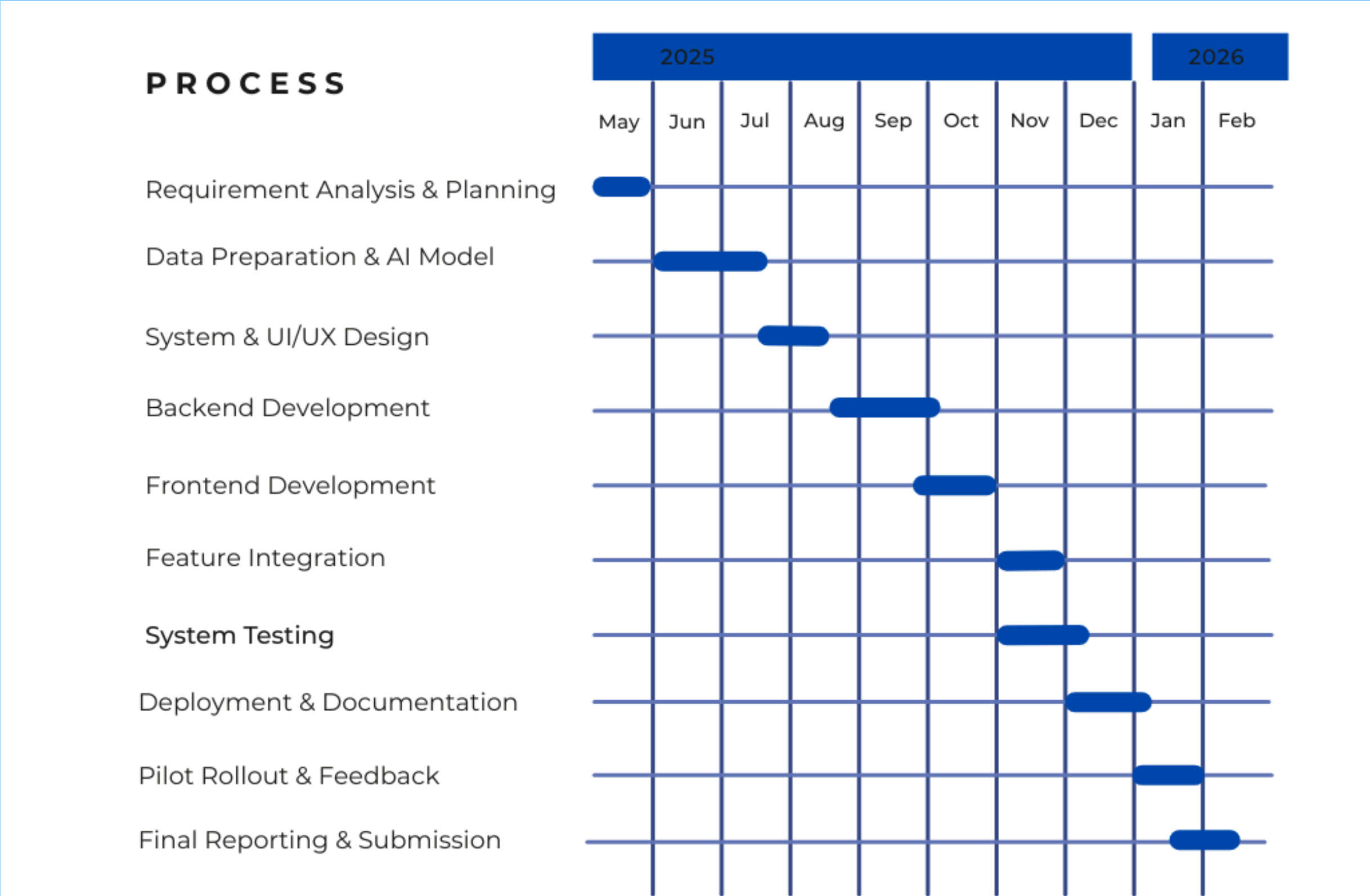
A5: RICH PICTURE — LOST & FOUND SYSTEM



A6: STAKEHOLDER ONION MODEL



A7: PROJECT GANTT CHART



A stylized, light blue line drawing of a human brain, positioned in the upper right corner of the slide. It is rendered in a simplified, almost abstract manner with visible gyri and sulci.

THANK YOU!